Software Requirements Specification (SRS)

AI-Powered Content Management System (AICMS)

Version: 1.0 | Date: 09-10-2026 | Status: Stage 1 – Proposed design / planning

Team / Author: Shreeverdhan M | SRN: PES1UG24AM265
Sripada Krishna Sameer Aditya | SRN: PES1UG24AM285
Srivamshi Chikkireddi | SRN: PES1UG24AM287
Shamanth SK | SRN: PES1UG24AM258

Submission boundary: This submission defines requirements, proposed design, and planned verification. It does not claim that a working implementation exists or that tests have been executed.

Revision History

Version	Date	Author	Change summary	Approval
1.0	09-10-2026	AICMS Team (Shreeverdhan M, Sripada Krishna Sameer Aditya, Srivamshi Chikkireddi, Shamanth SK)	Initial Stage 1 SRS for AICMS	Pending

Approvals

Role	Name	Signature / Email	Date
Course Coordinator			
Team Representative			

Table of Contents

- 1. Introduction
- 2. Overall description
- 3. External interface requirements
- 4. System features (detailed)
- 5. Non-functional requirements (detailed)
- 5.1 Security objectives and requirements
- 6. Quality attributes and acceptance tests
- 7. System models and diagrams
- 8. Requirements Traceability Matrix (RTM)

1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) defines the proposed requirements, interfaces, acceptance criteria, and verification references for the AI-Powered Content Management System (AICMS).

1.2 Scope

AICMS is a document organization and discovery layer. It accepts supported documents, extracts text and available metadata, suggests categories, extracts keywords/topics, creates semantic embeddings, indexes searchable content, and supports natural-language search. Authorized users can review and correct AI-generated metadata. The system does not replace a source-of-truth repository or perform unrelated content authoring in this Stage 1 scope.

1.3 Audience

Project evaluators, end users, content editors, administrators, security reviewers, developers, testers, and future maintainers.

1.4 Definitions

Term	Definition
AICMS	AI-Powered Content Management System.
Semantic search	Retrieval based on meaning/context in addition to exact keyword matches.
Embedding	A numeric vector representation of text used for semantic comparison.
OCR	Optical Character Recognition for text extraction from scanned pages.
RBAC	Role-Based Access Control.
API	Application Programming Interface.
RTM	Requirements Traceability Matrix.
PII	Personally Identifiable Information.
TLS	Transport Layer Security.
WCAG	Web Content Accessibility Guidelines.
p90	90th percentile: 90% of measured values are at or below this value.
STP	Software Test Plan.

2. Overall Description

2.1 Product Perspective

AICMS is proposed as a browser-based system with a server-side API, file/object storage, metadata database, document-processing workers, AI enrichment services, and a search index. The high-level pipeline is upload and validation → extraction → AI enrichment → metadata/index update → permission-filtered semantic search. Processing is asynchronous to accommodate variable parser and model latency.

2.2 Major Product Functions (Detailed)

- Upload and validate supported documents; detect probable duplicates.
- Extract document text and available metadata; optionally use OCR for scans.
- Suggest categories and extract keywords/topics; generate document embeddings.
- Index eligible text and metadata for semantic retrieval.
- Accept natural-language queries and return relevance-ranked results.
- Filter results by approved metadata fields.
- Allow authorized editors to correct AI-generated metadata.
- Enforce authentication and document-level authorization.
- Maintain audit events, processing states, operational metrics, taxonomy administration, and re-index jobs.
- Support deletion from active indexes in accordance with a future-approved retention policy.

2.3 User Roles and Characteristics

Role	Needs and permitted activities
General user	Upload permitted documents, search accessible documents, view authorized result details.
Content editor	Review and correct AI-generated categories, keywords, topics, and allowed metadata.
Administrator	Manage roles and taxonomy, request re-indexing, inspect system metrics and audit records.
Auditor / evaluator	Review approved audit/traceability evidence without unauthorized content modification.

2.4 Operating Environment

A modern browser communicates with a secured API. The server-side environment comprises a relational metadata store, protected document storage, background job execution, document parsing/OCR adapters, an approved model/embedding service, a search index, and monitoring/logging. Deployment may be institution-approved cloud or on-premises; the final target is not yet selected.

2.5 Constraints and Assumptions

- Only defined and tested file types, languages, and upload-size limits will be supported; the proposed initial scope is PDF, DOCX and TXT files up to 20 MB, English language.
- AI-generated labels are suggestions, not guaranteed facts; authorized human correction is supported.
- Search quality depends on extraction quality, language, model selection, and the evaluation corpus.
- Document permission checks must occur before returning result titles, snippets, embeddings, or content.
- Any external model service must be approved for the data classification and retention policy.

3. External Interface Requirements

3.1 User Interfaces

- Upload view: supported formats, progress, processing state, and safe validation errors.
- Search view: natural-language query, filters, relevance-ranked results, snippet, category, and clear no-results state.
- Document detail/editor view: available metadata and editable AI suggestions gated by role.
- Administration view: user/role management, taxonomy, re-index job status, audit, and aggregated metrics.
- Accessibility: keyboard navigation, visible focus, clear labels, adequate contrast, and understandable error/empty states.

3.2 Hardware Interfaces

No dedicated end-user hardware is required beyond a browser-capable device. Optional OCR or model acceleration hardware is a deployment choice and is not an assumed Stage 1 dependency.

3.3 Software Interfaces

- Document parser and optional OCR adapter for agreed file types.
- Approved model interface for categorization, keyword/topic extraction, and text/query embeddings.
- Metadata database and protected file/object storage.
- Search/index service supporting vector similarity and metadata filters.
- Optional institution identity provider.
- Logging, metrics, alerting, and background-job queue interfaces.

3.4 Communications

Browser/API and service-to-service traffic shall use HTTPS/TLS 1.2 or higher. Protected APIs require authentication and authorization, validate inputs, apply suitable request limits, and use sanitized errors. Logs shall not contain credentials or unnecessary raw document content.

4. System Features (Detailed)

Each functional requirement uses a stable AICMS-F identifier and includes a priority, stakeholder/source, measurable acceptance criterion, test-case reference, and dependency note.

4.1 Document Intake and Validation

Req ID	Requirement (shall...)	Type	Priority	Source / Stakeholder	Acceptance criteria / Test case reference	Comments / Dependencies
AICMS-F-001	The system shall allow an authenticated user to upload a supported document.	Functional	High	Content users	A valid supported file is accepted and assigned a unique document ID. Test: TC-UP-01	Requires authenticated session and storage.
AICMS-F-002	The system shall reject unsupported file types and files exceeding the configured size limit.	Functional	High	Security / operations	The file is rejected before processing and a safe explanatory message is displayed. Test: TC-UP-02	Allowed types and size limit must be configured.
AICMS-F-003	The system shall detect probable duplicate uploads using a configured content fingerprint or metadata policy.	Functional	Medium	Content users	Duplicate content triggers the configured warning or duplicate-handling action. Test: TC-UP-03	Duplicate policy is a configuration decision.
AICMS-F-004	The system shall show each uploaded document's processing status and any safe, actionable failure reason.	Functional	Medium	Content users	Queued, processing, completed, and failed states are distinguishable. Test: TC-UP-04	Avoid exposing internal stack traces.

4.2 Text Extraction and AI Enrichment

Req ID	Requirement (shall...)	Type	Priority	Source / Stakeholder	Acceptance criteria / Test case reference	Comments / Dependencies
AICMS-F-005	The system shall extract searchable text from supported text-based documents.	Functional	High	Content discovery	Text is linked to the correct document and can be indexed. Test: TC-EXT-01	Depends on supported format parsers.
AICMS-F-006	The system shall extract available document metadata such as title, author, creation date, and file type.	Functional	Medium	Content discovery	Available metadata is stored; unavailable fields remain empty rather than being invented. Test: TC-EXT-02	Metadata availability varies by format.
AICMS-F-007	The system shall route scanned documents through OCR when OCR is enabled for the file type.	Functional	Medium	Content discovery	OCR text is stored and uncertain/poor extraction is marked for review. Test: TC-EXT-03	OCR may be an optional integration.
AICMS-F-008	The system shall suggest one or more categories for a processable document.	Functional	High	Content editors	A category suggestion and model/version or review information is stored. Test: TC-AI-01	Suggestions are not guaranteed to be correct.
AICMS-F-009	The system shall extract representative keywords and topics from document text.	Functional	High	Content editors	Keywords/topics are stored for documents with sufficient extractable text. Test: TC-AI-02	Language/model coverage must be evaluated.

Req ID	Requirement (shall...)	Type	Priority	Source / Stakeholder	Acceptance criteria / Test case reference	Comments / Dependencies
AICMS-F-010	The system shall generate semantic embeddings for eligible document content.	Functional	High	Search users	Embedding creation succeeds or a visible retryable/failed processing state is recorded. Test: TC-AI-03	Depends on approved embedding model/service.

4.3 Indexing and Semantic Search

Req ID	Requirement (shall...)	Type	Priority	Source / Stakeholder	Acceptance criteria / Test case reference	Comments / Dependencies
AICMS-F-011	The system shall index extracted text and approved metadata for search.	Functional	High	Search users	A successfully processed document becomes discoverable through the search interface. Test: TC-IDX-01	Index writes must be idempotent.
AICMS-F-012	The system shall accept natural-language queries for document discovery.	Functional	High	Search users	A valid query returns authorized results or an understandable no-results state. Test: TC-SR-01	Depends on availability of search index.
AICMS-F-013	The system shall order semantic search results using a defined relevance-scoring strategy.	Functional	High	Search users	Relevant seeded documents rank above unrelated items in the agreed test set. Test: TC-SR-02	Ranking quality target requires benchmark corpus.
AICMS-F-014	The system shall support filtering search results by supported metadata fields, including category and file type.	Functional	Medium	Search users	Returned results satisfy all selected filter conditions. Test: TC-SR-03	Filterable fields must be defined.
AICMS-F-015	The system shall show authorized result titles, relevant snippets, categories, and available metadata.	Functional	Medium	Search users	Only permitted fields appear and empty/missing metadata is handled gracefully. Test: TC-SR-04	Authorization must be applied before formatting results.

4.4 Metadata, Permissions, and Audit

Req ID	Requirement (shall...)	Type	Priority	Source / Stakeholder	Acceptance criteria / Test case reference	Comments / Dependencies
AICMS-F-016	The system shall allow authorized content editors to correct AI-generated categories, keywords, and topics.	Functional	Medium	Content editors	Changes persist and are reflected in subsequent details/search indexing. Test: TC-MD-01	Edits require role authorization and audit.
AICMS-F-017	The system shall enforce user- and document-level access permissions for reading, editing, and deleting content.	Functional	High	Data owners	Unauthorized requests reveal no document body, title, snippet, or protected metadata. Test: TC-SEC-01; TC-SEC-05	Permission filtering applies to direct IDs and search.
AICMS-F-018	The system shall audit document access and administrative actions with actor, timestamp, action, and outcome.	Functional	High	Administrators / auditors	Representative actions create searchable audit events without logging credentials or raw secrets. Test: TC-SEC-02	Retention duration is policy-dependent.
AICMS-F-023	The system shall allow a registered user to log in and log out with credentials (or the institution identity provider) and shall establish an authenticated session.	Functional	High	All users	Valid credentials create a session; invalid credentials are rejected with a generic error; logout invalidates the session. Test: TC-SEC-09	Credential storage and lock-out per AICMS-SR-007 and SR-008.

4.5 Administration, Re-indexing, Deletion, and Monitoring

Req ID	Requirement (shall…)	Type	Priority	Source / Stakeholder	Acceptance criteria / Test case reference	Comments / Dependencies
AICMS-F-019	The system shall allow an authorized administrator to manage the approved category taxonomy.	Functional	Medium	Administrators	Only administrators can create, edit, or deactivate categories; changes are validated and audited. Test: TC-ADM-01	Taxonomy changes may trigger re-indexing.
AICMS-F-020	The system shall allow an authorized administrator to request re-indexing after taxonomy or model configuration changes.	Functional	Medium	Administrators	A re-index job is queued with visible status and a traceable job identifier. Test: TC-ADM-02	Requires background job processing.
AICMS-F-021	The system shall remove a deleted document from searchable results and associated indexes, subject to approved retention policy.	Functional	High	Data owners	Deleted content is not returned by search after deletion completes, and the event is audited. Test: TC-DATA-01	Backups/retention exceptions must be defined.
AICMS-F-022	The system shall expose basic operational metrics for upload failures, processing latency, search latency, and errors.	Functional	Medium	Administrators	Authorized administrators can inspect aggregated metrics without sensitive document content. Test: TC-OPS-02	Requires monitoring integration.
AICMS-F-024	The system shall allow an authorized administrator to create and deactivate user accounts and assign roles.	Functional	High	Administrators	Only administrators can change accounts or roles; changes take effect on the user's next request and are audited. Test: TC-ADM-03	Underpins RBAC (AICMS-SR-001); initial admin account seeded at deployment.

5. Non-functional Requirements (Detailed)

The following targets are proposed acceptance targets for a later test environment. The availability target requires an operating period to measure and is not asserted as achieved.

Req ID	Requirement	Category	Priority	Acceptance criteria / Measurement
AICMS-NF-001	For the agreed benchmark corpus and normal test load, 90% of search requests shall return results within 3 seconds.	Performance	High	p90 search latency is ≤ 3 s in the documented test environment. Test: TC-PERF-01
AICMS-NF-002	The proposed service shall target at least 99.5% monthly availability, excluding approved maintenance.	Reliability	Medium	Availability is calculable from monitoring records over a representative operating period. Test: TC-OPS-03
AICMS-NF-003	All browser-to-service and service-to-service network communications shall use HTTPS/TLS 1.2 or higher.	Security	High	Transport configuration review confirms modern TLS is enforced. Test: TC-SEC-03
AICMS-NF-004	Document parsing and AI enrichment shall run asynchronously so long-running jobs do not block unrelated upload or search requests.	Scalability	High	A long processing job is queued while independent API requests remain responsive under the agreed load. Test: TC-PERF-02
AICMS-NF-005	Core workflows shall support keyboard navigation, visible focus, accessible labels, and adequate contrast consistent with WCAG 2.1 AA where applicable.	Accessibility	Medium	Accessibility review finds no critical barrier in upload, search, and metadata-edit flows. Test: TC-UX-01

Req ID	Requirement	Category	Priority	Acceptance criteria / Measurement
AICMS-NF-006	Logs shall include timestamps and correlation IDs but shall exclude passwords, authentication tokens, and unnecessary raw document content.	Privacy / maintainability	High	Representative logs support diagnosis without secrets or excess content. Test: TC-SEC-06
AICMS-NF-007	Document records, metadata, and search indexes shall remain consistent after successful update and deletion operations.	Data integrity	High	Consistency checks find no stale deleted entries or untracked orphan records. Test: TC-DATA-02
AICMS-NF-008	The system shall clearly identify failed AI processing and shall not silently represent incomplete enrichment as successful.	Reliability	High	Simulated model failure produces a failed/retryable state and operational event. Test: TC-OPS-01
AICMS-NF-009	For the agreed labelled test set, at least 80% of documents shall receive a correct top category suggestion (matching the reviewed label).	Accuracy / AI quality	High	Accuracy >= 80% on the held-out labelled test set. Test: TC-AI-04
AICMS-NF-010	Automatic processing of a 10-page text-based document (extraction, categorization, keywords/topics, embedding, indexing) shall complete within 60 seconds under normal test load.	Performance	Medium	Median and p90 processing time <= 60 s in the named environment. Test: TC-PERF-03

5.1 Security Objectives and Requirements

5.1.1 Security Objectives

1. Confidentiality: only authenticated, authorized users may discover, preview, download, edit, or delete a document and its protected metadata.
2. Integrity and accountability: metadata, indexes, permissions, and administrative changes shall be protected from unauthorized modification and critical actions shall be auditable.
3. Availability: the system shall resist abusive input and request patterns (oversized or malicious uploads, repeated invalid logins, request flooding) so that legitimate users can continue to upload and search.

5.1.2 Security Requirements

Req ID	Requirement (shall...)	Type	Priority	Acceptance criteria / Test case reference
AICMS-SR-001	The system shall authenticate users and enforce least-privilege, role-based, document-level authorization for each protected operation.	Security – Access control	High	Unauthorized read/edit/delete and search attempts are denied. Test: TC-SEC-01
AICMS-SR-002	The system shall validate uploaded content and isolate parsers/OCR components when handling untrusted files.	Security – Input safety	High	Malformed, disguised, or disallowed files are rejected or safely contained. Test: TC-SEC-04
AICMS-SR-003	The system shall encrypt network traffic using HTTPS/TLS 1.2 or higher and shall disable insecure transport paths.	Security – Transport security	High	Configuration scan confirms secure transport; insecure requests are rejected or redirected safely. Test: TC-SEC-03
AICMS-SR-004	The system shall not disclose restricted document IDs, names, snippets, metadata, embeddings, or content in search results.	Security – Information protection	High	Cross-role tests reveal no restricted document information. Test: TC-SEC-05

Req ID	Requirement (shall...)	Type	Priority	Acceptance criteria / Test case reference
AICMS-SR-005	The system shall store secrets in approved secret/configuration storage and shall not emit them in logs or client errors.	Security – Secrets management	High	Secret/configuration and error/log inspection reveals no credential leakage. Test: TC-SEC-06
AICMS-SR-006	The system shall record permission changes, taxonomy changes, metadata edits, and document deletion in an auditable event trail.	Security – Auditability	Medium	Each action produces an event with actor, time, action, and outcome. Test: TC-SEC-02
AICMS-SR-007	The system shall store user passwords only as salted hashes (e.g., bcrypt or Argon2) and shall never display or log them.	Security – Credential protection	High	User store contains no plaintext passwords; passwords are absent from logs and responses. Test: TC-SEC-07
AICMS-SR-008	The system shall expire inactive sessions after 30 minutes and temporarily lock an account after 5 consecutive failed login attempts.	Security – Session protection	Medium	Expired session requires re-login; the 6th attempt after five failures is blocked and audited. Test: TC-SEC-08

6. Quality Attributes and Acceptance Tests

Quality attribute	Verification approach	Proposed acceptance / exit criterion
Functional completeness	Run mapped functional cases from the test register.	All high-priority requirements pass; failures are logged and triaged.
Search relevance	Evaluate a reviewed query-document benchmark.	Evaluate about 20–50 reviewed queries; target: a relevant document in the top 5 results for at least 80% of queries. Report Precision@5 and relevant-document coverage.
Performance	Run a documented search workload and concurrent processing workload.	AICMS-NF-001 and AICMS-NF-004 meet the targets in the named environment.
Security / privacy	Run role-bypass, upload-safety, transport, secret/log, and restricted-snippet tests.	No unresolved critical/high confidentiality or access-control defect at release gate.
Accessibility / usability	Keyboard-only checks and accessibility review of upload/search/edit flows.	No critical blockers; documented issues are remediated or formally accepted.
Data integrity	Update and delete documents; reconcile database metadata against search index.	No deleted document appears in search after completed deletion; reconciliation exceptions are handled.

Acceptance exit criteria: high-priority functional and security requirements are verified; no critical defects remain open; non-functional test results are recorded; and the RTM contains a disposition for every in-scope requirement. These are future release criteria, not current test results.

7. System Models and Diagrams

7.1 UML Use-Case Diagram 1 — General User and Content Editor

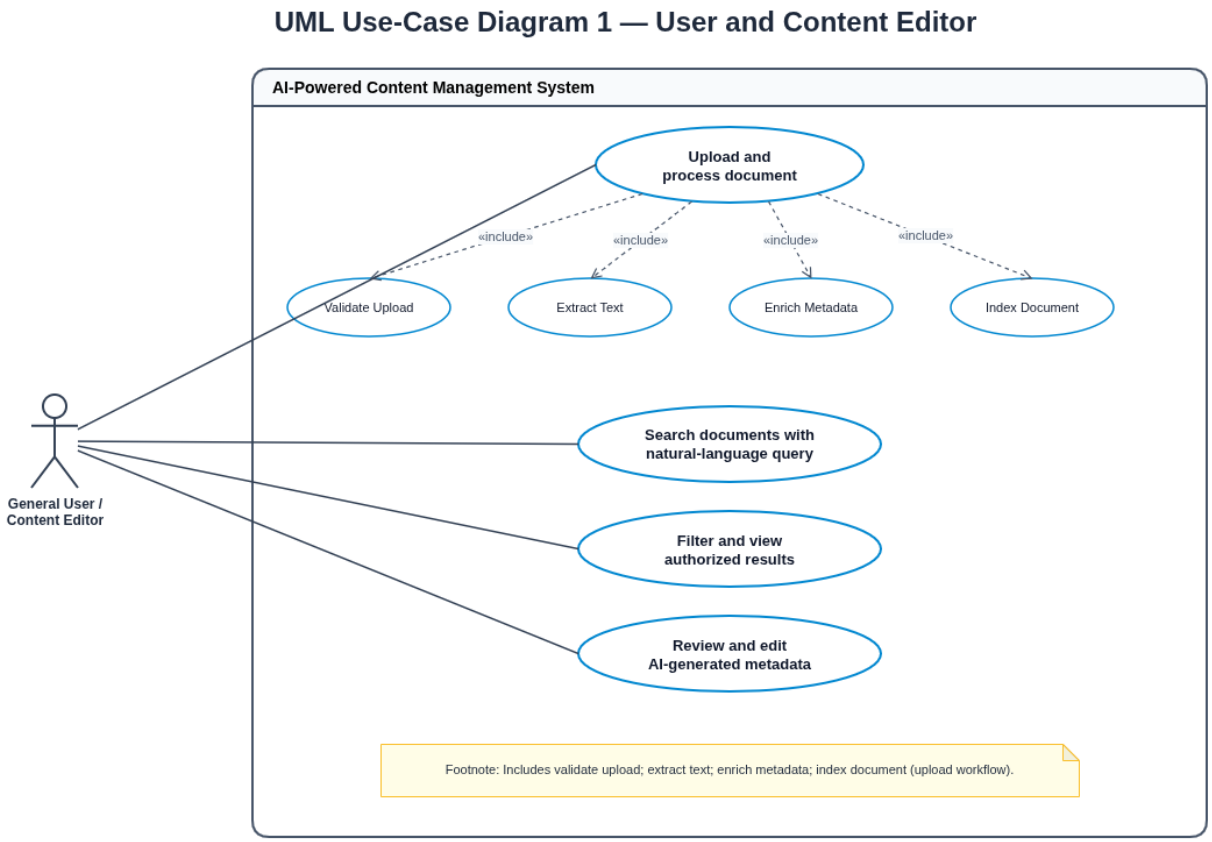


Figure 1. User/editor use cases within the AICMS system boundary.

7.2 UML Use-Case Diagram 2 — Administrator

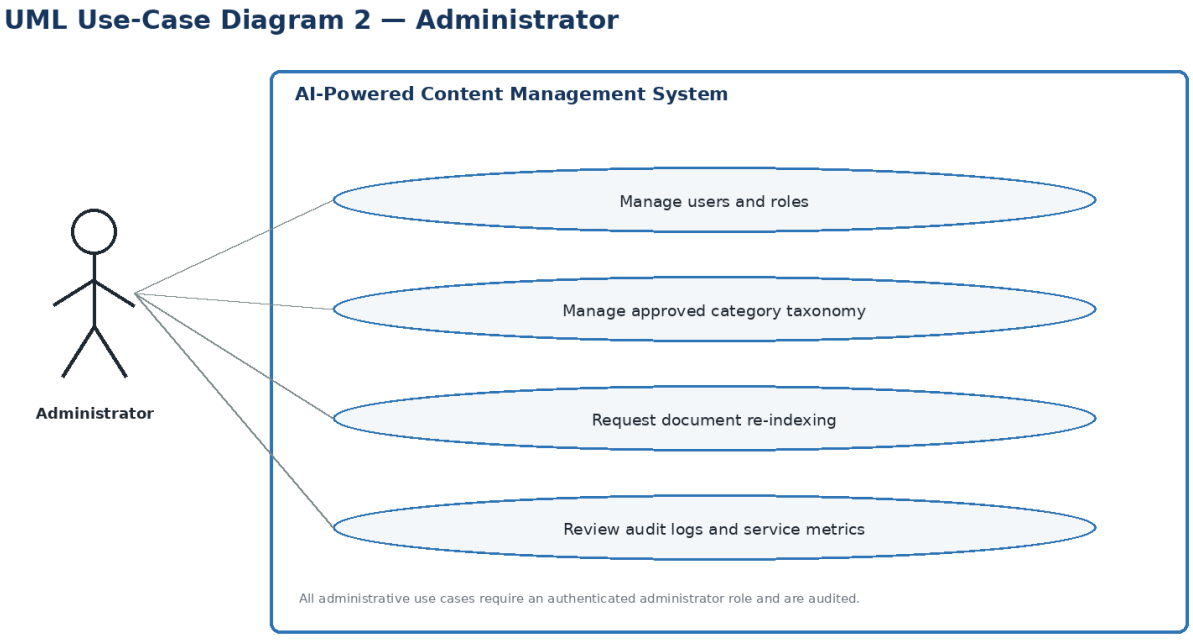


Figure 2. Administration use cases; privileged actions require administrator authorization.

7.3 Use-Case Notes

Use case	Precondition	Main success path	Alternative / exception
UC-01 Upload and process	User is authenticated and has upload permission.	Select file → validate → store → queue processing → extract/enrich/index → show status.	Unsupported file, duplicate policy, parser/model failure, or permission failure is shown safely.
UC-02 Search documents	User has an authenticated session if the repository is protected.	Enter natural-language query → embed/retrieve → apply permission filters → rank → display authorized snippets.	No results, index outage, timeout, or empty query produces a clear response.
UC-03 Review metadata	Editor has permission for the target document.	Review categories/keywords/topics → edit → validate → save → update index/audit.	Unauthorized edit is denied; re-index failure is visible.
UC-04 Administer system	Administrator is authenticated with required privileges.	Manage roles/taxonomy → request re-index → inspect audit/metrics.	Invalid changes are rejected; each administrative action is audited.

8. Requirements Traceability Matrix (RTM)

Status legend: N = Not run / planned; P = test passed; A = formally accepted by the project authority. All entries are currently N because no implementation or test execution is in scope for Stage 1.

Req ID	Requirement short	Section ref / Design Spec	Module	Test case(s)	Status (N/P/A)	Comments
AICMS-F-001	Upload supported document	4.1 / SAD §3.4	Web UI; Application API; Document Intake	TC-UP-01	N	Planned; Requires authenticated session and storage.
AICMS-F-002	Reject invalid files	4.1 / SAD §3.4	Document Intake	TC-UP-02	N	Planned; Allowed types and size limit must be configured.
AICMS-F-003	Detect duplicate uploads	4.1 / SAD §3.4	Document Intake; Metadata Service / DB	TC-UP-03	N	Planned; Duplicate policy is a configuration decision.
AICMS-F-004	Show processing status	4.1 / SAD §3.4	Web UI; Async Queue / Worker	TC-UP-04	N	Planned; Avoid exposing internal stack traces.
AICMS-F-005	Extract document text	4.2 / SAD §3.4	Text Extraction / OCR	TC-EXT-01	N	Planned; Depends on supported format parsers.

Req ID	Requirement short	Section ref / Design Spec	Module	Test case(s)	Status (N/P/A)	Comments
AICMS-F-006	Extract document metadata	4.2 / SAD §3.4	Text Extraction / OCR; Metadata Service / DB	TC-EXT-02	N	Planned; Metadata availability varies by format.
AICMS-F-007	OCR for scanned files	4.2 / SAD §3.4	Text Extraction / OCR	TC-EXT-03	N	Planned; OCR may be an optional integration.
AICMS-F-008	Suggest categories	4.2 / SAD §3.4	AI Enrichment Pipeline	TC-AI-01; TC-AI-04	N	Planned; Suggestions are not guaranteed to be correct.
AICMS-F-009	Extract keywords and topics	4.2 / SAD §3.4	AI Enrichment Pipeline	TC-AI-02	N	Planned; Language/model coverage must be evaluated.
AICMS-F-010	Generate embeddings	4.2 / SAD §3.4	AI Enrichment Pipeline; Search Index	TC-AI-03	N	Planned; Depends on approved embedding model/service.
AICMS-F-011	Index text and metadata	4.3 / SAD §3.4	Search Index; Async Queue / Worker	TC-IDX-01	N	Planned; Index writes must be idempotent.
AICMS-F-012	Natural-language query	4.3 / SAD §3.4	Search / Retrieval API; Web UI	TC-SR-01	N	Planned; Depends on availability of search index.
AICMS-F-013	Rank results by relevance	4.3 / SAD §3.4	Search / Retrieval API; Search Index	TC-SR-02	N	Planned; Ranking quality target requires benchmark corpus.
AICMS-F-014	Filter search results	4.3 / SAD §3.4	Search / Retrieval API	TC-SR-03	N	Planned; Filterable fields must be defined.
AICMS-F-015	Show authorized result details	4.3 / SAD §3.4	Search / Retrieval API; Web UI	TC-SR-04	N	Planned; Authorization must be applied before formatting results.
AICMS-F-016	Correct AI-generated metadata	4.4 / SAD §3.4	Metadata Service / DB; Web UI	TC-MD-01	N	Planned; Edits require role authorization and audit.
AICMS-F-017	Document-level access control	4.4 / SAD §3.4	Application API (RBAC); Metadata Service / DB	TC-SEC-01; TC-SEC-05	N	Planned; Permission filtering applies to direct IDs and search.
AICMS-F-018	Audit access and admin actions	4.4 / SAD §3.4	Audit / Observability	TC-SEC-02	N	Planned; Retention duration is policy-dependent.
AICMS-F-023	Login / logout and session	4.4 / SAD §3.4	Web UI; Application API (authentication)	TC-SEC-09	N	Planned; Credential rules per SR-007/SR-008.
AICMS-F-019	Manage category taxonomy	4.5 / SAD §3.4	Admin UI; Metadata Service / DB	TC-ADM-01	N	Planned; Taxonomy changes may trigger re-indexing.
AICMS-F-020	Request re-indexing	4.5 / SAD §3.4	Admin UI; Async Queue / Worker; Search Index	TC-ADM-02	N	Planned; Requires background job processing.
AICMS-F-021	Remove deleted documents from index	4.5 / SAD §3.4	Metadata Service / DB; Search Index; Async Queue / Worker	TC-DATA-01	N	Planned; Backups/retention exceptions must be defined.
AICMS-F-022	Operational metrics	4.5 / SAD §3.4	Audit / Observability	TC-OPS-02	N	Planned; Requires monitoring integration.
AICMS-F-024	Manage users and roles	4.5 / SAD §3.4	Admin UI; Application API (RBAC); Metadata Service / DB	TC-ADM-03	N	Planned; Underpins RBAC.
AICMS-NF-001	Search p90 response time <= 3 seconds	5 / SAD §3.4	Search / Retrieval API; Search Index	TC-PERF-01	N	Planned verification
AICMS-NF-002	Monthly availability target >= 99.5%	5 / SAD §3.4	Whole system; Audit / Observability	TC-OPS-03	N	Planned verification
AICMS-NF-003	HTTPS/TLS 1.2+ for network traffic	5 / SAD §3.4	Application API	TC-SEC-03	N	Planned verification
AICMS-NF-004	Asynchronous processing for long-running jobs	5 / SAD §3.4	Async Queue / Worker	TC-PERF-02	N	Planned verification
AICMS-NF-005	Accessible keyboard, focus, labels and contrast	5 / SAD §3.4	Web UI	TC-UX-01	N	Planned verification
AICMS-NF-006	Sanitized logs with timestamps and correlation IDs	5 / SAD §3.4	Audit / Observability	TC-SEC-06	N	Planned verification

Req ID	Requirement short	Section ref / Design Spec	Module	Test case(s)	Status (N/P/A)	Comments
AICMS-NF-007	Consistency across records, metadata and index	5 / SAD §3.4	Metadata Service / DB; Search Index	TC-DATA-01; TC-DATA-02	N	Planned verification
AICMS-NF-008	Visible/retryable AI failure; no false success	5 / SAD §3.4	AI Enrichment Pipeline; Async Queue / Worker	TC-AI-03; TC-OPS-01; TC-UP-04	N	Planned verification
AICMS-NF-009	≥ 80% categorization accuracy	5 / SAD §3.4	AI Enrichment Pipeline	TC-AI-04	N	Planned verification
AICMS-NF-010	Processing ≤ 60 s per 10-page document	5 / SAD §3.4	Async Queue / Worker; AI Enrichment Pipeline	TC-PERF-03	N	Planned verification
AICMS-SR-001	Authentication and least-privilege document-level RBAC	5.1 / SAD §3.9	Application API (RBAC)	TC-ADM-01; TC-ADM-03; TC-SEC-01; TC-SEC-09	N	Planned verification
AICMS-SR-002	Validate uploads and isolate parsers/OCR	5.1 / SAD §3.9	Document Intake; Text Extraction / OCR	TC-SEC-04; TC-UP-02	N	Planned verification
AICMS-SR-003	Encrypt traffic using HTTPS/TLS 1.2+	5.1 / SAD §3.9	Application API; network configuration	TC-SEC-03	N	Planned verification
AICMS-SR-004	No restricted data in results or snippets	5.1 / SAD §3.9	Search / Retrieval API	TC-SEC-05; TC-SR-04	N	Planned verification
AICMS-SR-005	Secure secret storage; exclude secrets from logs	5.1 / SAD §3.9	Configuration / secrets; Audit / Observability	TC-SEC-06	N	Planned verification
AICMS-SR-006	Audit permission, taxonomy, metadata and deletion changes	5.1 / SAD §3.9	Audit / Observability	TC-ADM-01; TC-MD-01; TC-SEC-02	N	Planned verification
AICMS-SR-007	Salted password hashing	5.1 / SAD §3.9	Application API (authentication)	TC-SEC-07	N	Planned verification
AICMS-SR-008	Session expiry and account lock-out	5.1 / SAD §3.9	Application API (authentication)	TC-SEC-08	N	Planned verification

Appendix A — References

- ISO/IEC/IEEE 29148, requirements engineering (reference framework).
- OWASP Application Security Verification Standard and File Upload guidance.
- W3C Web Content Accessibility Guidelines (WCAG) 2.1.
- Selected parser, OCR, model, database, and search documentation to be confirmed when technology decisions are made.